Sec: SR.IIT_*CO-SC(MODEL-A& APEX) GTM- 20 Date: 18.01.2025
Time: 3 HRS JEE-MAIN Max. Marks: 300

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	A	C	B	A	A	D	A	D	D
11	12	13	14	15	16	17	18	19	20
C	B	D	D	C	D	C	D	C	D
21	22	23	24	25					
150	1	2	3	10					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	A	A	B	B	D	A	A	C	C
36	37	38	39	40	41	42	43	44	45
C	A	B	B	C	C	B	A	D	C
46	47	48	49	50					
6	15	827	17	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	D	A	B	A	A	D	A	C	C
61	62	63	64	65	66	67	68	69	70
A	D	D	B	B	C	B	A	D	B
71	72	73	74	75					
832	130	100	385	6006					

PHYSICS

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1. Such a material is a diamagnetic material, that shows magnetisation induced against the external magnetic field. This magnetisation causes the internal magnetic field to decrease. Hence, $B_{in} < B_{ext}$.
2. The magnetic field is increased by the relative permeability of the material. But the magnetic intensity doesn't change.
3. In process I, temperature changes but volume remains constant, so mean free path is unaffected. In process II, volume increases adiabatically but number of molecules and molecular diameter remain unchanged. Hence, number of particles per unit volume decreases and the mean free path increases as per the equation: $\lambda_{mean} = \frac{1}{\sqrt{2}n\pi d^2}$. Here, $n =$ number of particles per unit volume.
4. Ground-Frame Momentum (p_G):

$$p_G = \text{mass} \times \text{velocity} = 2.00\text{kg} \times 2.000\text{m/s} = 4.00\text{kgm/s}$$

Observer-Frame Momentum (p_O):

$$p_O = \text{mass} \times \text{relative velocity} = 2.00\text{kg} \times (2.000\text{m/s} - 1.50\text{m/s})$$

$$\Rightarrow \quad p_O = 2.00\text{kg} \times 0.50\text{m/s} = 1.0\text{kgm/s}$$

5. Yield strength is a limiting stress of a material, which is force per unit area. Its formula is $\frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$.
6. Conceptual
7. The angle made by liquid's free surface with wet surface of solid is obtuse.
8. During the day, land heats faster, causing air above it to rise and cooler air from the water to move toward the land.
9. This describes energy conservation (first law), not the second law.
10. CO molecules have 3 translational, 2 rotational, and 2 vibrational mode. $C_v = \left(\frac{3+2+2}{2}\right)R = 3.5R$.
11. The energy for each quadratic term is $\frac{1}{2}k_BT$.
12. Electric field is zero in the connecting wires, but is present in the resistance from higher potential towards lower potential side. Electric field lines are exiting from surface Σ_1 and electric field lines are entering the surface Σ_2. Therefor positive and negative charges are present in them respectively.
13. This is related to air resistance and mass, not surface tension.
14. Speed of all the electromagnetic waves are identical in vacuum
15. Converging power is equal to the tangent of angle of deviation per unit height of the light ray

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16. Magnification = $\frac{f_{objective}}{f_{eyepiece}} = 100$, apparent separation = $100 \times 1' = 1.67^{\circ}$.
17. Reflecting telescopes use mirrors and avoid chromatic aberration unlike lenses.
18. Sky appears blue due to selective scattering of blue color of the sunlight.
19. Only transverse waves can show polarization
20. Conceptual
21. Using Curie's law:

$200\text{K} \times 0.0075 = 100\text{K} \times (x) \times 10^{-4}$

$\Rightarrow \quad x = 150$

22. The minima formed must be completely dark. This occurs when the intensity of two slits are the same.
23. $\frac{2\pi}{\lambda} = 3.14 \times 10^{3}$

$\Rightarrow \quad 2 \times 10^{-3}\text{m} = \lambda$

24. For each direction of vibration (x, y and z), there are two degrees of freedom. So, a total of 6 degrees of freedom. So molar heat capacity is $C_P = C_V = \frac{6}{2}R = 3R$
25. Relative velocity of one train $w.r.t.$ other $w.r.t.$ $= 10 + 10 = 20m/s$

Relative acceleration $= 0.3 + 0.2 = 0.5m/s^2$ $0.3 + 0.2 = 0.5m/s^2$

If train crosses each other then from $s = ut + \frac{1}{2}at^2$ $s = ut + \frac{1}{2}at^2$

$As, s = s_1 + s_2 = 100 + 125 = 225$

$\Rightarrow 225 = 20t + \frac{1}{2} \times 0.5 \times 0.5 \times t^2 \Rightarrow 0.5t^2 + 40t - 450 = 0$

$\Rightarrow t = \frac{-40 \pm \sqrt{1600 + 4.(005) \times 450}}{1} = -40 \pm 50$

$\therefore t = 10sec \therefore t = 10sec$ (Taking +ve value)

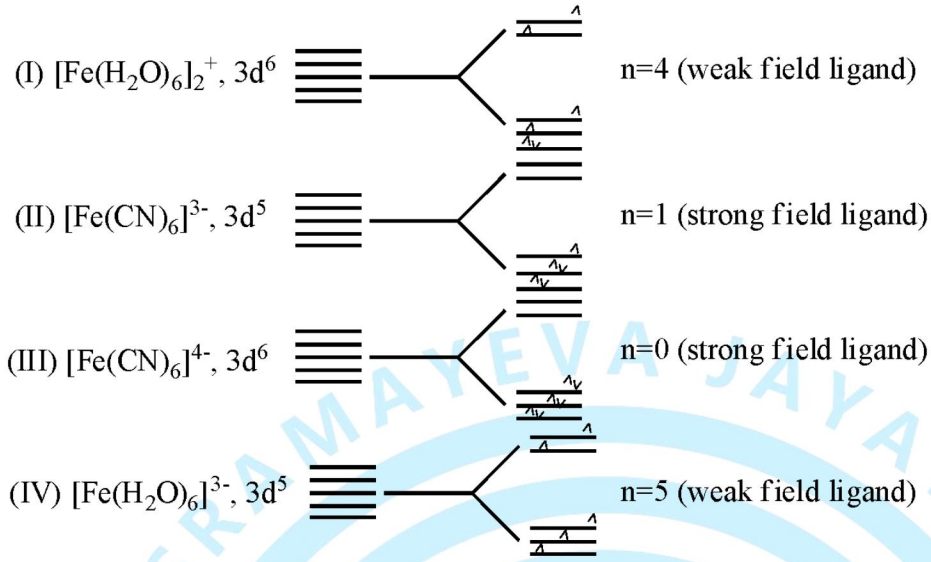
CHEMISTRY

26. **S_1:** It is because Mn^{2+} has $3d^5$ configuration which has extra stability.
S_2: Not titanium but Copper, because with +1 oxidation state an extra stable configuration, $3d^{10}$ results.
S_3: It is not stable as it undergoes disproportionation; $2Cu^{+}(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$. The E^{o} value for this is favourable.
S_4: Much larger third ionisation energy of Mn (where the required change is d^5 to d^4) is mainly responsible for this.

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27. It is an amphoteric in nature and it reacts with alkalies as well as with acids to give $V_4O_9^{2-}$ and VO^{2+} respectively.

28.



29. $Cu^{2+} + I^- \rightarrow Cu^+ + I_2$

$n_{eq} Na_2S_2O_3 = n_{eq} I_2 = n_{eq} Cu^{2+}$

Or $\frac{V \times 0.4}{1000} \times 1 = \frac{50 \times 0.2}{1000} \times 1 \Rightarrow V_{Na_2S_2O_3} = 25\,ml$

30. $A \longrightarrow B + 2C + 3D$

Optical rotation - θ_1 θ_2 θ_3

$t = 0$ a 0 0 0 Optical rotation at t = 0 is zero.

$t = t$ $a - x$ x 2x 3x Optical rotation at time= t

$(x.\theta_1 + 2x.\theta_2 + 3x.\theta_3)$

$t = \infty$ 0 a 2a 3a Optical rotation at time = ∞ is

$(a.\theta_1 + 2a.\theta_2 + 3a.\theta_3)$

$(x.\theta_1 + 2x.\theta_2 + 3x.\theta_3) = rt$; $(a.\theta_1 + 2a.\theta_2 + 3a.\theta_3) = r_\infty$

$x = \frac{r_t}{\theta_1 + 2\theta_2 + 3\theta_3}$; $a = \frac{r_\infty}{\theta_1 + 2\theta_2 + 3\theta_3}$

$k = \frac{1}{t} \ln\left[\frac{a}{a-x}\right] = \frac{1}{t} \ln\left[\frac{r_\infty}{r_\infty - r_t}\right]$

31. $2A + B \longrightarrow C$ rate $= k[A][B]$

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The value of k (velocity constant) is always independent of the concentration of reactant and it is a function of temperature only.

32. $H^+(aq)+OH^-(aq)\rightarrow H_2O(\ell)$

So $\Delta H^o=-573=-285.5-\Delta H_f^o\left(H^+,aq\right)-\Delta H_f{}^o\left(OH^-,aq\right)$

So, $\Delta H_f{}^o\left(OH^-,aq\right)=-228.5kJ/mole$ (as $\Delta H_f{}^o\left(H^+,aq\right)=0$)

33. $\Delta E=q+w$

$$W_{BC}=\frac{1}{2}\left(2V^o-V^o\right)\left(P^o-3P^o\right)+\left(2V^o-V^o\right)\left(0-P^o\right)=-2P^oV^o$$

$$\Delta E=nC_V\Delta T=1\times\frac{3}{2}R\left(\frac{P_o2V_o}{R}-\frac{3P_oV_o}{R}\right)=-\frac{3}{2}P^oV^o$$

$$q_{BC}=\Delta E-W=-\frac{3}{2}P^oV^o+2P^oV^o=\frac{1}{2}P^oV^o$$

35. $SrCO_3 \rightleftharpoons Sr^2+CO_3^{2-}$

$S_1 \quad S_1+S_2$

$ZnCO_3 \rightleftharpoons Zn^{2+}+CO_3^{2-}$

$S_2 \quad S_1+S_2$

$$S_1\times\left(S_1+S_2\right)=10^{-10}$$

$$S_2\times\left(S_1+S_2\right)=1.5\times10^{-11}$$

$$\Rightarrow\frac{S_1}{S_2}=\frac{\left[Sr^{2+}\right]}{\left[Zn^{2+}\right]}=\frac{10^{-10}}{1.5\times10^{-11}}=\frac{10}{1.5}=\frac{100}{15}=\frac{20}{3}$$

$$S_{pH=2}=\frac{10^{-38}}{\left(10^{-12}\right)^3}=10^{-2}M \qquad S_{pH=11}=\frac{10^{-38}}{\left(10^{-3}\right)^3}=10^{-29}M$$

36. (A) (B)

37. Order of solubility : Complex formation > Pure water > Common ion effect

AgBr form complex $\left[Ag\left(NH_3\right)_2\right]^+$ in NH_3, so solubility is maximum in NH_3 (aq).

39. $n_{eq}KHC_2O_4=n_{eq}KMnO_4$

$$\frac{x}{M}\times2=\frac{100\times0.02}{1000}\times5$$

(1)

$n_{eq}KHC_2O_4=n_{eq}Ca\left(OH\right)_2$

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$$\frac{y}{M}\times 1=\frac{100\times 0.5}{1000}\times 2 \quad (2)$$

$$\therefore \frac{x}{y}=\frac{1}{2}$$

46. (1) All lanthanides exhibits +3 O.N. but few also exhibit +2 and +4 O.N.
(2) Correct
(3) Incorrect as all belongs to 6th period. (4) True due to lanthanide contraction.
(5) Incorrect as basicing decreases from $Ce(OH)_3$ to $Lu(OH)_3$ becomes of fazans rule.
(6) Correct
(7) Correct
(8) All lanthanides have –ve S.R.P. hence all releases hydrogen.

47. $\left[Rh(en)_2(NO_2)_2\right]NO_3$

Linkage	Geo.	Optical
NO_2 NO_2	Cis + trans	Cis
NO_2 ONO	Cis + trans	Cis
ONO ONO	Cis + trans	Cis
NO_2 NO_3	Cis + trans	Cis
ONO NO_3	Cis + trans	Cis

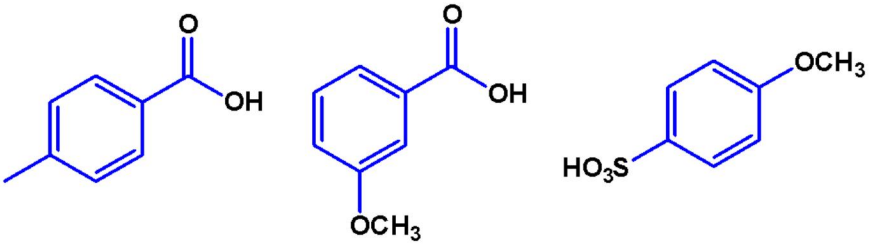
$\left[Rh(en)_2(NO_2)NO_3\right]NO_2$

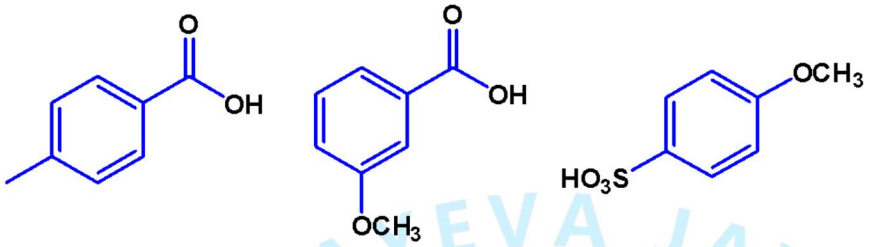
$5\times 3=15$ isomers

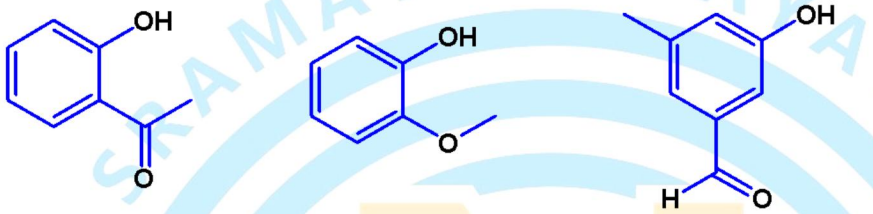
48. Equation (vi) can be generated by the following manipulations.
Eq. (i) – Eq. (ii) – Eq. (iii) – Eq. (iv) – ½ Eq. (v)
Carrying out the corresponding manipulations on ΔH^0_L,
we get $\Delta H^0_L = (-563 - 419 - 88 + 322 - 79)$ kJ mol^- $= -827$ kJ/mole

49.

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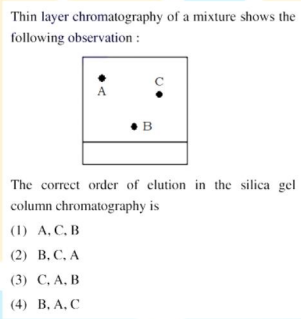
p 

q 



r all except third row second one

50. Thin layer chromatography of a mixture shows the following observation



The correct order of elution in the silica gel column chromatography is
1) A, C, B
2) B, C, A
3) C, A, B
4) B, A, C

MATHEMATICS

51. B and C will contain three digit numbers of the form $9k+2$ and $9k+\ell$ respectively. We need to find sum of all elements in the set $B\cup C$ effectively. Now , $S(B\cup C)=S(B)+S(C)-S(B\cap C)$ where $S(k)$ denotes sum of elements of set k.

Also $B=\{101,109,....992\}$

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$\therefore \quad S(B)=\frac{100}{2}(101+992)=54650$

Case-I : If $\ell=2$ then $B\cap C=B$

$\therefore S(B\cup C)=S(B)$

Which is not possible as given sum is $274\times 400=109600$.

Case-II : If $\ell\neq 2$ then $B\cap C=\phi$

$\therefore S(B\cup C)=S(B)+S(C)=400\times 274$

$\Rightarrow 54650+\sum_{k=11}^{110}9k+\ell=109600$

$\Rightarrow 9\sum_{k=11}^{110}k+\sum_{k=11}^{110}\ell=54950$

$\Rightarrow 9\left(\frac{100}{2}(11+110)\right)+\ell(100)=54950$

$\Rightarrow 54450+100\ell=54950\Rightarrow \ell=5$

52. Given a set $\{1,2,...50\}$ Possible choices of P are $2,3,5,7,11,13,17,23,29,31,37,41,43,\text{and}\,47$. So, we can calculate no. of elements in R_1 as $(2,2^o),(2,2^1).(2,2^5)$

$(3,3^o),....(3,3^3)$

$(5,5^o),....(5,5^2)$

$(7,7^o),...(7,7^2)$

$(11,11^o),....(11,11^1)$, Similarly for 13,17,23,29,31,37,41,43 and 47. Every number of P^n should lie in the given set $\{1,2,...50\}$. Total number of relations are $n(R_1)=6+4+3+3+(2\times 10)=36$ Similarly for relation R_2, $(2,2^o),(2,2^1)....(47,47^o),(47,47^1)$

Total number of relations are $n(R_2)=2\times 14=28$ Required difference $n(R_1)-n(R_2)=36-28=8$

53. Set $S=\{1,2,3,...12\}$

$A\cup B\cup C=S, A\cap B=B\cap C=A\cap C=\phi$

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$\therefore$ Each sets contain 4 elements.

$\therefore$ The number of ways to partition $= {}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4$

$= \frac{12!}{4!8!} \times \frac{8!}{4!4!} \times \frac{4!}{4!0!} = \frac{12!}{(4!)^3}$

54. The general term is

$T_{k+1} = {}^{n}C_k (x)^{\frac{n-k}{2}} (-6)^k (x)^{\frac{-3}{2}k}$

$\because$ for constant term

$\frac{n-k}{2} - \frac{3}{2}k = 0 \quad \Rightarrow n - 4k = 0 \Rightarrow k = n/4$

Now, sum of remaining term is

$(-5)^n - \left({}^{n}C_{\frac{n}{4}} (-6)^{\frac{n}{4}} \right) = 649$

By observation $(625 + 24 = 649)$, we get $n = 4$

$\because$ $n = 4 \text{ and } k = 1$ And, $\alpha = {}^{4}C_1(-6) = -24$

Required is coefficient of x^{-4} in $\left(\sqrt{x} - \frac{6}{x^{3/2}} \right)^4$ is

${}^{4}C_1(-6)^3 = \lambda\alpha \Rightarrow \alpha = 36$

55. Let $P = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2.3} + \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{2^2.3} + \frac{1}{2.3^2} - \frac{1}{3^3}\right) + ...$

$\Rightarrow P\left(\frac{1}{2} + \frac{1}{3}\right) = \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \left(\frac{1}{2^3} + \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{3^4}\right) + ...$

$\Rightarrow \frac{5P}{6} = \frac{\frac{1}{4}}{1 - \frac{1}{2}} - \frac{\frac{1}{9}}{1 + \frac{1}{3}} \Rightarrow \frac{5P}{6} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}$

$\therefore P = \frac{1}{2} = \frac{\alpha}{\beta} \qquad \therefore \alpha = 1, \beta = 2$

So, $\alpha + 3\beta = 7$

56. $\sum f_i = 62 \Rightarrow 3k^2 + 4k - 2 = 62$

$\Rightarrow 3k^2 + 16k - 12k - 64 = 0 \quad \Rightarrow k = 4 \text{ or } -\frac{16}{3} (\text{rejected})$

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Now, $\mu = \dfrac{\sum f_i x_i}{\sum f_i}$ $[\because \text{put } k = 4]$

$$\mu = \frac{8+2(15)+3(15)+4(17)+5}{62} = \frac{156}{62} \quad \ldots.(i)$$

As, $\sigma^2 = \sum f_i x_i^2 - \left(\sum f_i x_i\right)^2$

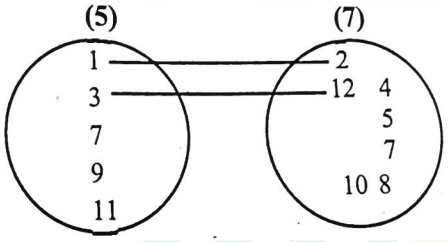
$$= \left(8\times1+15\times4+15\times9+17\times16+1\times25\right) - \left(\frac{156}{62}\right)^2$$

$$\sigma^2 = \frac{500}{62} - \left(\frac{156}{62}\right)^2 \quad \ldots..(ii)$$

$$\sigma^2 + \mu^2 = \frac{500}{62} \Rightarrow \left[\sigma^2 + \mu^2\right] = 8$$

By (i) and (ii)

57.



$A = \{1,3,7,9,11\}, \quad B = \{2,4,5,7,8,10,12\}$

$f(1) + f(3) = 14$

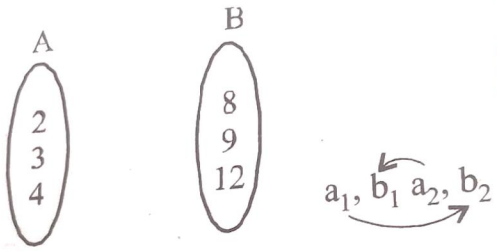
Case (i) 2 + 12

Case (ii) 4 + 10

Total number of one-one mapping

$$2\times\left(2\times5\times4\times3\right) = 240$$

58.



a_1 divides b_2

Each elements has 2 choices $\Rightarrow 3\times2 = 6$

a_2 divides b_1

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Each elements has 2 choices $\Rightarrow 3\times 2=6$

Total = $6\times 6=36$

59. Domain and codomain = $\{1,2,3,....,20\}$. There are five multiple of 4 as 4,8,12,16 and 20. And there are 6 multiple of 3 as 3,6,9,12,15,18. Since, when ever k is multiple of 4 then f(k) is multiple of 3 then total number of arrangement = $={}^{6}C_{5}\times 5!=6!$
Remaining 15 elements can be arranged in 15! Ways. Since, for every input, there is an output $\Rightarrow$ function f (k) is onto

$\therefore$ Total number of arrangements = $15!.6!$

60. $f(A)=|A|=ad-bc=15$

$\Rightarrow (ad,bc)=(9,-6)$ or $(6,-9)$

Case-I: If ad = 9 and bc = –6 then
All possible ways of ad = (3, 3), (–3, –3)
And all possible ways of bc = (2, –3) , (–2, 3), (3, –2), (–3, 2)

$\therefore$ Total ways = $2\times 4=8$

Case- II: If ad = 6 and bc = –9 then
All possible ways of ad = (3, 2), (–3, –2), (2, 3), (–2, –3)
And all possible ways of bc = (3, –3), (–3, 3)
Total ways = 8 $\Rightarrow$ Total such matrices are = 16

61. Given, 4m + n = 22
17m + 4n = 93
After solving (i) & (ii), we get.
m = 5, n = 2

$\Rightarrow |A|=m-n=3$

Now, $|2adj(adj5A)|=2^{5}|5A|^{16}=2^{5}.5^{80}|A|^{16}$

$=2^{5}.5^{80}.3^{16}=3^{11}.5^{80}.6^{5}$

$\Rightarrow a+b+c=11+80+5=96$

62. Given, $f(x)=[x(x-1)]+\{x\}$

$f(0^{+})=-1+0=-1,$ $f(1^{+})=0+0=0$

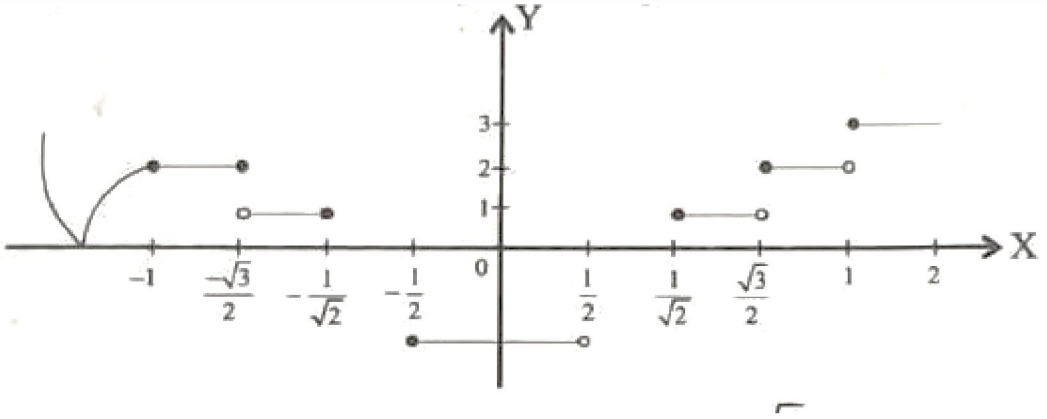
$f(0)=0,$ $f(1)=0$

$f(0")=0+1=1$ $f(1")=-1+1=0$

$\therefore f(x)$ is continuous at x = 1, discontinuous at x = 0

63. Let's draw the graph of the given function.

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Here, there are 7 points $\pm\frac{1}{2},\pm\frac{1}{\sqrt{2}},\pm\frac{\sqrt{3}}{2}$ and 1 at which function is discontinuous.

64. $f(x)=\frac{1}{\sqrt{2x^3-9x^2+12x+4}}$

$$f'(x)=\frac{-1}{2}\left(\frac{(6x^2-18x+12)}{(2x^3-9x^2-12x+4)^{3/2}}\right)$$

$$=\frac{-6(x-1)(x-2)}{2(2x^3-9x^2+12x+4)^{3/2}};\ f(1)=\frac{1}{3}\quad f(2)=\frac{1}{\sqrt{8}}$$

and

It is increasing function $\frac{1}{3}<I<\frac{1}{\sqrt{8}}\Rightarrow\frac{1}{9}<I^2<\frac{1}{8}$

65. According to the question Slope of line = –9

$$\Rightarrow 27r_1+\frac{9r_2}{2}=-9 \quad \text{..... (i)}$$

$$\lim_{x\to 3}\frac{\int_3^x 8t^2dt}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}$$

$$=\lim_{x\to 3}\frac{8x^2}{\frac{3r_2^2}{2}-2r_2x-3r_1x^2-3}$$

(Using L Hospital's)

$$=\frac{72}{\frac{3r_2}{2}-6r_2-27r_1-3}$$

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$$= \frac{72}{-\frac{9r_2}{2} - 27r_1 - 3} \quad \text{[Using (i)]}$$

$$= \frac{72}{9-3} = 12$$

66.

$$\int dy = \int \frac{\left(2x^2+2x+3\right)}{x^4+2x^3+3x^2+2x+2}dx$$

$$y = \int \frac{\left(2x^2+2x+3\right)}{\left(x^2+1\right)\left(x^2+2x+2\right)}dx$$

$$y = \int \frac{dx}{x^2+2x+2} + \int \frac{dx}{x^2+1}$$

$$y = \tan^{-1}\left(x+1\right) + \tan^{-1}x + C$$

$$y(-1) = \frac{-\pi}{4}; \frac{-\pi}{4} = 0 - \frac{\pi}{4} + C \Rightarrow C = 0$$

$$\Rightarrow y(x) = \tan^{-1}\left(x+1\right) + \tan^{-1}x$$

$$y(0) = \tan^{-1}1 = \frac{\pi}{4}.$$

67.

$$\frac{dy}{dx} + \frac{\sqrt{2}}{2\cos^4 x - \cos 2x}y = xe^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$$

Here given differential equation is in the form of $\frac{dy}{dx} + py = Q$

$$\int \frac{dx}{2\cos^4 x - \cos 2x} \left\{ \because P = \int \frac{dx}{2\cos^4 x - \cos 2x} \right\}$$

$$\int \frac{dx}{\cos^4 x + \sin^4 x} = \int \frac{\operatorname{cosec}^4 x dx}{1+\cot^4 x}$$

$$= -\int \frac{t^2+1}{t^4+1}dt = -\int \frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}dt = \frac{-1}{\sqrt{2}}\tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)$$

Cotx = t

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$$=-\frac{1}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\cot 2x\right) IF = e^{-\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$$

$$ye^{-\tan^{-1}\left(\sqrt{2}\cot 2x\right)} = \int x dx; ye^{-\tan^{-1}\left(\sqrt{2}\cot 2x\right)} = \frac{x^2}{2} + c$$

$$y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} + c \Rightarrow c = 0; y = \frac{x^2}{2} e^{\tan^{-1}\left(\sqrt{2}\cot 2x\right)}$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18} e^{\tan^{-1}\left(\sqrt{2}\cot\frac{2\pi}{3}\right)} = \frac{\pi^2}{18} e^{-\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)}$$

$$\alpha = \sqrt{\frac{2}{3}} \Rightarrow 3\alpha^2 = 2$$

68. Given, $S_n = 3 + 7 + 11 + ...n$ terms

$$= \frac{n}{2}\left(6 + (n-1)4\right) = 3n + 2n^2 - 2n \qquad \{a = 6, d = 4\}$$

$$= 2n^2 + n$$

Now, $\sum_{k=1}^{n} S_k = 2\sum_{k=1}^{n} K^2 + \sum_{k=1}^{n} K$

$$= 2, \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)(4n+5)}{6}$$

Since, $$40 < \frac{6}{n(n+1)} \sum_{k=1}^{n} S_k < 42 \Rightarrow 40 < 4n + 5 < 42$$

$$\Rightarrow 35 < 4n < 37 \Rightarrow n = 9$$

69. $\sum x_1 = 15 \times 20 = 300$ ..... (i)

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \quad \text{..... (ii)}$$

$$\sum x_1^2 = 234 \times 20 = 4680$$

$$\frac{\sum (x_1 + a)^2}{20} = 178 \Rightarrow \sum (x_1 + \alpha)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2\alpha \sum x_1 + \sum \alpha^2 = 3560$$

$$\Rightarrow 4680 + 600\alpha + 20\alpha^2 = 3560 \Rightarrow \alpha^2 + 30\alpha + 56 = 0$$

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$\Rightarrow (\alpha+28)(\alpha+2)=0 \Rightarrow a=-2,-28$

Square of maximum value of α is 4.

70. The desired functions will contain either one element or two elements in its codomain of which '2' always belongs to f(A).

71. $A=\{1,2,3,.....100\}$ and $B-C=\{3k+1: k\in \text{even}\} \Rightarrow B-C=\{7,13,19,.....,97\}$ So,

$A\cap(B-C)=\{7,13,19,.....97\}$ Sum of all elements $=\frac{16}{2}[7+97]=832$

72. Given curves are, $x=\sin\left(2\tan^{-1}\alpha\right)=\frac{2\alpha}{1+\alpha^2}$ and

$y=\sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)=\sin\left(\sin^{-1}\frac{1}{\sqrt{5}}\right)=\frac{1}{\sqrt{5}}$ We have set with relation $y^2=1-x$

$\frac{1}{5}=1-\frac{2\alpha}{1+\alpha^2}$ {from value of x & y}

$\Rightarrow 1+\alpha^2=5+5\alpha^2-10\alpha \Rightarrow 2\alpha^2-5\alpha+2=0$

So, $\alpha=2,\frac{1}{2}$

Take, $\sum_{\alpha\in S}16\alpha^3=16\times 2^3+16\times\frac{1}{2^3}=130$

73. Given $X=\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}$ $X^2=\begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix}$ thus

$Y=\begin{bmatrix}\alpha&\beta&\gamma\\0&\alpha&\beta\\0&0&\alpha\end{bmatrix}, Z=\begin{bmatrix}\alpha^2&-\alpha\beta&\beta^2-\alpha\gamma\\0&\alpha^2&-\alpha\beta\\0&0&\alpha^2\end{bmatrix}$

$Y.Y^{-1}=I$

$$\begin{bmatrix}\alpha&\beta&\gamma\\0&\alpha&\beta\\0&0&\alpha\end{bmatrix}\begin{bmatrix}\frac{1}{5}&\frac{-2}{5}&\frac{1}{5}\\0&\frac{1}{5}&\frac{-2}{5}\\0&0&\frac{1}{5}\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

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$$\frac{\alpha}{5}=1\Rightarrow\alpha=5,-\frac{2}{5}\alpha+\frac{\beta}{5}=0\Rightarrow\beta=10$$

$$\frac{\alpha}{5}-\frac{2\beta}{5}+\frac{\gamma}{5}=0\Rightarrow\gamma=15$$

$$\Rightarrow(\alpha-\beta+\gamma)^2=(5-10+15)^2=100$$

74. Given function is $f(x)=[x]-10$

$$\int_0^{10} f(x).dx=-10-9-8-....-1$$

Take, $=-\frac{10-11}{2}=-55$

Take,

$$\int_0^{10}(f(x))^2\,dx=10^2+9^2+8^2+....+1=\frac{10-11}{2}=55$$

$=-55+385+55=385$

75. $\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$

$$T_{r+1}={}^{15}C_r\left(t^2x^{\frac{1}{5}}\right)^{15-r}.\frac{(1-x)^{\frac{r}{10}}}{t^r}$$

For independent of t,

$r+2r-30=0$ So r = 10

So, maximum value of ${}^{15}C_{10}x(1-x)$ will be at $x=\frac{1}{2}$

$3003\times\frac{1}{2}\times\frac{1}{2}=\frac{3003}{2};8k=6006.$

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